

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

CSE 3102, Lab No. 10 , Lab Work

Task 1: Database and Table Creation

Create a database called “MusicStreamDB”

```
CREATE DATABASE MusicStreamDB;
```

Create a table titled “Artists” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
artist_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
artist_name	VARCHAR (100)	NOT NULL
country	VARCHAR (50)	NOT NULL
debut_year	INT	NOT NULL
genre	VARCHAR (50)	NOT NULL

```
CREATE TABLE Artists (  
  artist_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
  artist_name VARCHAR(100) NOT NULL,  
  country VARCHAR(50) NOT NULL,  
  debut_year INT NOT NULL,  
  genre VARCHAR(50) NOT NULL  
);
```

Create a table titled “Songs” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
song_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
title	VARCHAR (100)	NOT NULL
release_date	DATE	NOT NULL
duration_seconds	INT	NOT NULL
artist_id	INT	NOT NULL

```
CREATE TABLE Songs (  
  song_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
  title VARCHAR(100) NOT NULL,  
  release_date DATE NOT NULL,
```

```

duration_seconds INT NOT NULL,
artist_id INT NOT NULL,
FOREIGN KEY (artist_id) REFERENCES Artists(artist_id)
);

```

Create a table titled “Users” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
user_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
user_name	VARCHAR (100)	NOT NULL
email	VARCHAR (100)	NOT NULL
country	VARCHAR (50)	NOT NULL
date_joined	DATE	NOT NULL

```

CREATE TABLE Users (
    user_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
    user_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL,
    country VARCHAR(50) NOT NULL,
    date_joined DATE NOT NULL
);

```

Create a table titled “Streams” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
stream_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
user_id	INT	NOT NULL
song_id	INT	NOT NULL
stream_date	DATE	NOT NULL
play_duration	INT	NOT NULL

```

CREATE TABLE Streams (
    stream_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
    user_id INT NOT NULL,
    song_id INT NOT NULL,
    stream_date DATE NOT NULL,
    play_duration INT NOT NULL,
    FOREIGN KEY (user_id) REFERENCES Users(user_id),
    FOREIGN KEY (song_id) REFERENCES Songs(song_id)
);

```

Task 2: Data Insertion

Populate the table “Artists” with the following data.

artist_name	country	debut_year	genre
Taylor Swift	USA	2006	Pop
Ed Sheeran	UK	2011	Pop
Arijit Singh	India	2013	Bollywood
BTS	South Korea	2013	K-pop
Adele	UK	2008	Soul

```
INSERT INTO Artists (artist_name, country, debut_year, genre) VALUES
('Taylor Swift', 'USA', 2006, 'Pop'),
('Ed Sheeran', 'UK', 2011, 'Pop'),
('Arijit Singh', 'India', 2013, 'Bollywood'),
('BTS', 'South Korea', 2013, 'K-pop'),
('Adele', 'UK', 2008, 'Soul');
```

Populate the table “Songs” with the following data.

title	release_date	duration_seconds	artist_id
Love Story	2008-09-15	235	1
Perfect	2017-03-03	263	2
Tum hi ho	2013-04-16	270	3
Dynamite	2020-08-21	200	4
Hello	2015-10-23	295	5
Shape of You	2017-01-06	240	2

```
INSERT INTO Songs (title, release_date, duration_seconds, artist_id)
VALUES
('Love Story', '2008-09-15', 235, 1),
('Perfect', '2017-03-03', 263, 2),
('Tum hi ho', '2013-04-16', 270, 3),
('Dynamite', '2020-08-21', 200, 4),
('Hello', '2015-10-23', 295, 5),
('Shape of You', '2017-01-06', 240, 2);
```

Populate the table “Users” with the following data.

user_name	email	country	date_joined
Alice	alice@mail.com	USA	2022-01-05
Bob	bob@mail.com	UK	2022-02-12
Chaitanya	chaitanya@mail.com	India	2023-03-18
Diana	diana@mail.com	USA	2023-04-20

```
INSERT INTO Users (user_name, email, country, date_joined) VALUES
('Alice', 'alice@mail.com', 'USA', '2022-01-05'),
('Bob', 'bob@mail.com', 'UK', '2022-02-12'),
('Chaitanya', 'chaitanya@mail.com', 'India', '2023-03-18'),
('Diana', 'diana@mail.com', 'USA', '2023-04-20');
```

Populate the table “Streams” with the following data.

user_id	song_id	stream_date	play_duration
1	1	2023-06-01	235
1	6	2023-06-02	240
2	2	2023-06-03	263
3	3	2023-06-01	270
3	4	2023-06-05	180
4	5	2023-06-02	295
2	6	2023-06-06	240

```
INSERT INTO Streams (user_id, song_id, stream_date, play_duration)
VALUES
(1, 1, '2023-06-01', 235),
(1, 6, '2023-06-02', 240),
(2, 2, '2023-06-03', 263),
(3, 3, '2023-06-01', 270),
(3, 4, '2023-06-05', 180),
(4, 5, '2023-06-02', 295),
(2, 6, '2023-06-06', 240);
```

Task 3: SQL Queries

a) Query 1: Display all artists

```
SELECT * FROM Artists;
```

artist_id	artist_name	country	debut_year	genre
1	Taylor Swift	USA	2006	Pop
2	Ed Sheeran	UK	2011	Pop
3	Arijit Singh	India	2013	Bollywood
4	BTS	South Korea	2013	K-pop
5	Adele	UK	2008	Soul
6	Taylor Swift	USA	2006	Pop
7	Ed Sheeran	UK	2011	Pop
8	Arijit Singh	India	2013	Bollywood
9	BTS	South Korea	2013	K-pop
10	Adele	UK	2008	Soul

b) Query 2: List all song titles and duration

```
SELECT title, duration_seconds FROM Songs;
```

title	duration_seconds
Love Story	235
Perfect	263
Tum hi ho	270
Dynamite	200
Hello	295
Shape of You	240

c) Query 3: Find users who joined after 2023

```
SELECT user_name, date_joined FROM Users WHERE date_joined > '2023-01-01';
```

user_name	date_joined
Chaitanya	2023-03-18
Diana	2023-04-20

d) Query 4: Show streams where play duration > 240

```
SELECT user_id, song_id, play_duration FROM Streams WHERE  
play_duration > 240;
```

user_id	song_id	play_duration
2	2	263
3	3	270
4	5	295

e) Query 5: Count how many songs each artist has

```
SELECT a.artist_name, COUNT(s.song_id) AS song_count  
FROM Artists a  
JOIN Songs s ON a.artist_id = s.artist_id  
GROUP BY a.artist_name;
```

artist_name	song_count
Adele	1
Arijit Singh	1
BTS	1
Ed Sheeran	2
Taylor Swift	1

f) Query 6: Show total streams per user.

```
SELECT u.user_name, COUNT(st.stream_id) AS total_streams
FROM Users u
JOIN Streams st ON u.user_id = st.user_id
GROUP BY u.user_name;
```

user_name	total_streams
Alice	2
Bob	2
Chaitanya	2
Diana	1

g) Query 7: Find songs streamed more than once.

```
SELECT s.title, COUNT(st.stream_id) AS stream_count
FROM Songs s
JOIN Streams st ON s.song_id = st.song_id
GROUP BY s.title
HAVING COUNT(st.stream_id) > 1;
```

title	stream_count
Shape of You	2

h) Query 8: Find average play duration per user.

```
SELECT u.user_name, AVG(st.play_duration) AS avg_duration
FROM Users u
JOIN Streams st ON u.user_id = st.user_id
GROUP BY u.user_name;
```

user_name	avg_duration
Alice	237.5000
Bob	251.5000
Chaitanya	225.0000
Diana	295.0000

i) Query 9: Find the most streamed song.

```
SELECT s.title, COUNT(st.stream_id) AS total_streams
FROM Songs s
JOIN Streams st ON s.song_id = st.song_id
GROUP BY s.title
ORDER BY total_streams DESC
LIMIT 1;
```

title	total_streams
Shape of You	2

j) Query 10: Show total play time per artist.

```
SELECT a.artist_name, SUM(st.play_duration) AS total_play_time
FROM Artists a
JOIN Songs s ON a.artist_id = s.artist_id
JOIN Streams st ON s.song_id = st.song_id
GROUP BY a.artist_name;
```

artist_name	total_play_time
Adele	295
Arijit Singh	270
BTS	180
Ed Sheeran	743
Taylor Swift	235